

PatchCvT: Extended Mathematical Formulation

1 Input Representation

$$x \in \mathbb{R}^{H \times W \times 3} \quad (1)$$

2 CNN Feature Extraction

$$F = f_{\theta}(x) \quad (2)$$

$$F \in \mathbb{R}^{H' \times W' \times C} \quad (3)$$

$$Q = \{q_j\}_{j=1}^N \quad (4)$$

3 Memory Bank

$$M = \{m_k\}_{k=1}^K \quad (5)$$

4 Memory Retrieval

$$d_{jk} = \|q_j - m_k\|_2 \quad (6)$$

$$\alpha_{jk} = \frac{\exp(-d_{jk}/\tau)}{\sum_{l=1}^K \exp(-d_{jl}/\tau)} \quad (7)$$

$$\tilde{q}_j = \sum_{k=1}^K \alpha_{jk} m_k \quad (8)$$

5 Residual Degradation Signal

$$r_j = q_j - \tilde{q}_j \quad (9)$$

6 Activation Energy

$$a_j = \|F_j\|_2 \quad (10)$$

7 Semantic Token Construction

$$z_j = [q_j \| \tilde{q}_j \| r_j \| a_j] \quad (11)$$

8 Transformer Encoding

$$z'_j = \text{Transformer}(z_j) \quad (12)$$

9 Semantic Robustness

$$R_j = g_\phi(z'_j) \quad (13)$$

10 Semantic Preservation Gate

$$S_j = \sigma(W_r R_j - W_e \|r_j\|) \quad (14)$$

$$\hat{z}_j = S_j z'_j \quad (15)$$

11 Semantic Consistency Learning

Let x^+ denote a degraded version of the same object.

$$L_{\text{semantic}} = 1 - \cos(\hat{z}(x), \hat{z}(x^+)) \quad (16)$$

12 Memory Consistency Loss

$$L_{\text{memory}} = \frac{1}{N} \sum_{j=1}^N \|q_j - \tilde{q}_j\|^2 \quad (17)$$

13 Robustness Learning

$$R_j^{\text{target}} = \exp(-\|r_j\|) \quad (18)$$

$$L_{\text{robust}} = \frac{1}{N} \sum_{j=1}^N (R_j - R_j^{\text{target}})^2 \quad (19)$$

14 Gate Consistency

$$S_j^{\text{target}} = \exp(-\|r_j\|) \quad (20)$$

$$L_{\text{gate}} = \frac{1}{N} \sum_{j=1}^N |S_j - S_j^{\text{target}}| \quad (21)$$

15 Final Objective

$$L = \lambda_1 L_{\text{semantic}} + \lambda_2 L_{\text{memory}} + \lambda_3 L_{\text{robust}} + \lambda_4 L_{\text{gate}} \quad (22)$$

$$\lambda_1 = 1.0 \quad (23)$$

$$\lambda_2 = 0.5 \quad (24)$$

$$\lambda_3 = 0.25 \quad (25)$$

$$\lambda_4 = 0.25 \quad (26)$$

16 Interpretation

PatchCvT optimizes semantic retention under degradation rather than anomaly detection. The model preserves representations that remain semantically meaningful despite corruption, occlusion, dirt, blur, deformation, or partial destruction.